

Amnah M. Eltahir, PhD
DATA SCIENTIST
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Accomplished communicator adept at orchestrating seamless exchanges between diverse laboratories across universities, specializing in distilling complex research findings into compelling narratives for global audiences. Solid foundation in data science and postdoctoral research experience in advanced data analysis, combined with a unique blend of interdisciplinary skills to navigate collaborative environments comprising technicians, engineers, scientists, and clinicians. Proven record of accomplishment in managing and analyzing large datasets, coupled with a talent for teaching and presentation, poised to drive impactful communication strategies and facilitate transformative scientific discourse at every level.

KEY TECHNICAL SKILLS

Data Science:

- **Languages:** Python, SQL, MATLAB, Unix/Bash, R, LabVIEW
- Visualizing data using Python, MATLAB, and R to generate figures and reports
- Git version control
- Developing quality control software to assess large volume data
- Writing SQL queries for large, complex data sets
- Hypothesis development and testing
- Presenting results to key stakeholders

Machine Learning:

- Developing Python packages to handle large data sets
- Using MATLAB and Python tools to explore machine learning methods
- Python packages NumPy, Pandas, SciKit-learn, TensorFlow and PyTorch

AWARDS & HONORS

- **Neuroscience Scholars Program Fellow** 2019 - 2021
- **Physics Honor Society Inductee** - 2013

PROFESSIONAL EXPERIENCE

Data Scientist

Koneksa Health Inc.

May 2022 – April 2024

- Wrote prototype and production code to produce, process, and QC (quality control) large datasets (in-house and AWS cloud).
- Produced clinical datasets for client data transfers and analysis and presented reports.
- Effectively and proactively communicated with external and internal teams to gather information and contributed to data transfer specification contracts.
- Tabulated, visualized, and analyzed data, troubleshooted, resolved data quality issues, and own resolution of open items.

Postdoctoral Research Fellow

Addiction Institute of Mount Sinai

March 2021 – April 2022

- Acquire and manage imaging data collected at the Addiction Institute at Mount Sinai (AIMS).
- Build analysis pipelines for functional magnetic resonance imaging (fMRI) analysis.
- Manage off-site collaboration with the University of California, Los Angeles.
- Build fMRI analysis pipelines: general linear model (GLM) and dynamic causal modeling.
- Use REDCap databases to identify statistically relevant clinical variables to neuroimaging findings.

Postdoctoral Research Assistant

Virginia Tech

October 2020 – March 2021

- Guided new developments in voltammetry sensing protocols for human patient applications.

Graduate Research Assistant

Virginia Tech

August 2014 - September 2020

- Invented a new “low amplitude” sensing method for the detection of dopamine, serotonin, and norepinephrine.
- Developed Python and Jupyter Notebook code base to read voltammetry data files and prepare for machine learning applications.
- Performed machine learning and AI (Artificial Intelligence) analyses on voltammetry data:
- Designed in-vitro experiments to investigate new voltammetry sensing methods.
- Conducted in-vitro sample preparation and data collection.
- Applied Support Vector Regression analysis to study functional connectivity between deep and surface-level brain structures during resting state and task.
- Wrote MATLAB code to analyze functional near-infrared spectroscopy (fNIRS) data.
- Coordinated fNIRS study of infants receiving constraint-induced motor therapy (CIMT).
- Computed inversion recovery times of in-vitro and in-vivo MRI contrast agents.

EDUCATION

Ph.D. Biomedical Engineering | Virginia Polytechnic Institute and State University – September 2020

M.S. Biomedical Engineering | Virginia Polytechnic Institute and State University – May 2018

B.S. Physics | Virginia Polytechnic Institute and State University – December 2013